Evidence-Carrying World Models, Typed Governance Semantics, and Categorical Backends: A Conceptual Primer

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Abstract

This note consolidates a family of conceptual ideas developed around world-model calculus, evidence semantics, governance reasoning, and typed semantic backends. Its aim is didactic rather than archival: to state the cleanest current picture, separate what is already mathematically sharp from what remains conjectural, and record a stable set of design commitments for future formalization work.

The central thesis is that a world-model calculus should be understood first as an *evidence evaluator*, not as a global proof theory. The additive fragment—where query extraction is a monoid homomorphism from posterior-state revision into an evidence carrier—is mathematically distinguished and deserves to remain a named canonical subclass. More realistic semantics for provenance, correlation, overlap, forgetting, and locality should be layered *around* that additive core rather than replacing it. Modal, neighborhood, first-order, higher-order, and probabilistic relational semantics can then be hosted as semantic backends or extensional embeddings; governance and treaty reasoning become typed application layers over the same substrate.

The note also proposes a categorical placement: externally, as an *evidence-valued institution*; internally, as an *evidence-valued hyperdoctrine* or doctrine of typed queries. On the syntax side, dependent type theory and categories with families provide the natural candidate for the internal language of the trusted ontology/kernel layer, while presheaf-topos infrastructure remains the right ambient semantic home for the broader indexed semantics stack.

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1 Purpose and Scope

This document is a conceptual primer for a research line centered on:

- a *world-model calculus* for evidence-carrying semantic evaluation,
- typed governance and treaty semantics built on top of that calculus,
- categorical transport layers (institution, hyperdoctrine, Beck–Chevalley),
- modal, neighborhood, first-order, higher-order, and probabilistic relational backends,
- and the possibility that a dependent type theory supplies the internal language of the trusted ontology layer.

The note is deliberately *conceptual*. It is not a changelog, not a proof index, and not a full literature survey. It aims instead to preserve the strongest conceptual picture now available so that future development can resume from a stable basis.

2 The Core Idea: Evidence First, Views Second

The starting point is a simple but important methodological separation:

Probability is what one queries; evidence is what one carries.

The intended semantics is not that a world-model state directly *is* a probability distribution, a proof state, or a truth assignment. Rather, a world-model state carries enough information that queries can extract an *evidence object*, and then different operational views of that evidence can be selected: strength, confidence, acceptance, interval bounds, posterior parameters, and so on.

This separation matters for at least four reasons.

1. It keeps raw support distinct from acceptance policy.
2. It permits multiple views over the same carried evidence.
3. It prevents confidence, probability, and actuality from being conflated.
4. It makes semantic backends composable: multiple logics can target the same evidence layer while exposing different operational selectors.

2.1 The additive evaluator

The cleanest core interface is:

```
class WorldModel (State Query : Type*) [EvidenceType State] where
  evidence : State -> Query -> Evidence
  evidence_add : forall W1 W2 q,
    evidence (W1 + W2) q = evidence W1 q + evidence W2 q
```

The state type is equipped with revision/addition, and query extraction lands in an additive evidence carrier. This says that each query induces a monoid homomorphism from the revision algebra of states into the evidence algebra.

This is the mathematically distinguished *additive evidence ledger* fragment. It is not merely a convenient implementation choice. It captures exactly the semantics of independent sufficient-statistic aggregation in several canonical families.

3 Why the additive fragment is special

The additive fragment should remain a named privileged subclass even if the framework is generalized around it.

3.1 Ledger before policy

Exact additivity is the cleanest semantics of *accumulating raw evidence*: independent observations or independent posterior fragments add. This records how much support and counter-support have been gathered without yet deciding what to trust, what to discount, what to forget, or what to accept.

Trust discounting, overlap correction, anti-double-counting, and forgetting are all philosophically meaningful, but they are already *epistemic policy* choices. They should therefore be layered around additive accumulation rather than used to erase its identity.

3.2 Canonical sufficient-statistic semantics

The additive fragment is also singled out by conjugate Bayesian families. For example:

- Beta–Bernoulli: evidence counts are sufficient statistics.
- Dirichlet–multinomial: category counts are sufficient statistics.
- Normal–Gamma: count, sum, and sum-of-squares are sufficient statistics.

In each case, the sufficient statistics aggregate by addition. This makes the additive world-model fragment the canonical semantics of independent evidence accumulation for a broad and useful class of empirical models.

3.3 A universal-property viewpoint

The mathematically strongest current story is not that additive world models are the unique semantics of *all* evidence, but that they are the unique *free additive extension* of atomic observations.

Given atomic observation contributions

$$a : \mathrm{Obs} \to \mathrm{Query} \to Ev,$$

one obtains a unique additive extension

$$E : \mathrm{Multiset}\,\mathrm{Obs} \to \mathrm{Query} \to Ev$$

satisfying:

$$E(\{o\}, q) = a(o, q),$$
$$E(\sigma_1 + \sigma_2, q) = E(\sigma_1, q) + E(\sigma_2, q).$$

The right long-term theorem target is a full classification result:

> Every additive generic world model on multisets is uniquely determined by its singleton observation surface.

That would justify the additive layer as the canonical semantics of independent atomic evidence.

4 Why the generic shell is not vacuous

A recurring source of confusion is the intentional triviality of the bare state judgment.

4.1 Admissible states versus proof theory

The context-free world-model judgment should be read as:

> Any posterior state is admissible input to the evaluator.

This is *not* the claim that any proposition is derivable. Query judgments remain deterministic because for fixed state and query the evidence is fixed by the extractor. So the shell behaves more like big-step evaluation than like theoremhood.

The right rule of thumb is:

- the bare shell is an evaluator host,
- trusted derivability begins only in provenance-restricted or context-indexed layers,
- and proof-like content enters through certified rewrite rules, consequence rules, and closure operators.

4.2 Semantic nontriviality via endpoints

The framework becomes nontrivial through instantiated backends and endpoint correspondences.

For example, if one instantiates states as singleton pointed Kripke models or singleton pointed neighborhood models, then query strength recovers ordinary satisfaction truth values. With soundness and completeness assumptions for the external proof systems, implication-level proof-theoretic correspondences can be transported into world-model strength inequalities.

Thus the generic shell is not a deep global proof system, but it is a faithful semantic host for nontrivial proof-theoretic endpoints.

5 Views are selectors, not semantics

Strength, confidence, acceptance, and interval bounds should all be treated as *selectors* or *views* of evidence, not as the semantics itself.

5.1 No universal "better evidence" order

Raw evidence often comes with an obvious algebraic order, but that order need not coincide with "better evidence" for all purposes. In particular, an order that treats both positive and negative mass as monotone can make confidence increase while support quality decreases.

A more coherent approach is to define *view-induced preorders*. For example, with a strength context ctx and confidence parameter κ one may use:

$$e_1 \preceq_{ctx,\kappa} e_2 \iff \mathrm{strengthWith}(ctx, e_1) \leq \mathrm{strengthWith}(ctx, e_2) \wedge \mathrm{confidence}(\kappa, e_1) \leq \mathrm{confidence}(\kappa, e_2).$$

This keeps kernel algebra and acceptance policy conceptually distinct.

5.2 Actuality as evidence, not metaphysics

A closely related clarification concerns actuality. A "really exists" or `rexist` tag should not be read as metaphysical certainty. In an evidence-based world-model calculus, the correct reading is:

> actuality is an *event-occurrence evidence channel*, and categorical "actuality for governance" is a downstream acceptance policy.

This yields a clean two-layer design:

- `eventualityEvidence / eventEvidence`
- `acceptedOccurrence / eventAccepted`

The second is a policy-relative categorical layer; the first is the semantic one.

6 Subsumption, compliance, and treaty semantics

A major application layer developed around these ideas is a typed governance and treaty semantics.

6.1 Event subsumption

Practical governance does not only compare exact events. It often asks whether an abstract obligation is satisfied by a more specific concrete event.

The clean solution is a subsumption relation on eventualities:

- same predicate,
- same polarity,
- and every thematic role specified in the abstract event is matched by the concrete one.

This is the right formal replacement for brittle exact matching when checking whether an observed event fulfills an abstract clause.

6.2 Treaty kernels

On top of event subsumption, a minimal treaty kernel becomes natural:

- treaty clauses with obligor, obligee, modality, event pattern, optional deadline, and optional repair clause,
- timestamped treaty traces,
- separate predicates for *obligation fulfillment* and *prohibition violation*,
- and later policy-gated accepted traces.

The conceptual point is that compliance should not be post-hoc rule checking only. Treaty semantics should influence planning and repair. This is a useful minimal formal layer for an AI-to-AI treaty economy.

7 Modal and logical backends

The world-model calculus is broad enough to host several kinds of semantics.

7.1 Kripke semantics

A Kripke backend can be obtained by taking states to be multisets of pointed models and letting query evidence count supporting versus refuting points. Singleton states then recover ordinary pointed-model truth.

This gives a clean adequacy bridge:

> singleton query strength is ordinary modal truth; multisets yield evidential ensemble semantics over modal hypotheses.

7.2 Neighborhood semantics

Neighborhood semantics is even more native to evidence-style reasoning. It supports weaker and non-normal modalities, and it aligns naturally with families of evidence sets rather than only relational accessibility.

In the current conceptual picture:

- Kripke semantics is the clearest classical adequacy witness,
- neighborhood semantics is the better long-term evidence-native modal backend.

7.3 First-order and higher-order semantics

The same evaluator pattern can host set-theoretic FOL and Henkin-style HOL:

- states become multisets of pointed structures and assignments,
- queries become formulas in context,
- singleton adequacy recovers ordinary model satisfaction,
- and implication-level closure wrappers can transport proof-theoretic consequence into world-model strength inequalities.

This suggests that the world-model calculus is not tied to modal logic. It is better viewed as a semantics carrier for multiple logical families.

8 Probabilistic relational semantics: ProbLog and MLNs

ProbLog and Markov Logic Networks fit naturally at the extensional semantic level.

8.1 Weighted-world compilation

Both frameworks induce weighted possible-world semantics. One can compile them into a weighted-world world-model state and define evidence by splitting total world weight into support and refutation mass for a query.

Then query strength is the induced marginal probability of the query. In this sense:

> ProbLog and MLNs can be *subsumed extensionally* by the world-model calculus.

8.2 What is and is not preserved

This does *not* mean that the current additive state-revision operation is the native program-combination semantics of ProbLog or MLNs. Their intensional composition laws are global and non-additive in important ways.

So the correct claim is:

- exact query-level semantic subsumption: yes,
- exact native syntactic composition under additive revision: no.

The right architecture is therefore a weaker evaluator core with the additive world-model retained as the canonical independent-evidence specialization.

9 Categorical placement

The correct categorical home is not a single topos claim and not the claim that this calculus is itself the syntactic category of a theory.

9.1 Externally: evidence-valued institution

Externally, the cleanest formulation is as an *evidence-valued institution*:

- a category of signatures or contexts,
- sentences/queries over each signature,
- models/world-model states over each signature,
- and evidence-valued satisfaction stable under change of notation.

This captures the idea that the calculus is a semantic target for many logical front ends.

9.2 Internally: evidence-valued hyperdoctrine

Internally, the strongest picture is hyperdoctrinal:

- typed queries over a categorical base,
- reindexing along substitutions or morphisms,
- existential and universal transport,
- Beck–Chevalley coherence.

This is exactly the right setting for first-order and typed query semantics.

9.3 Ambient home: presheaf-topos flavored semantics

Presheaf topoi are the right *ambient* semantic home for much of the stack: strong enough to support internal intuitionistic logic, predicate fibrations, change of base, and Yoneda-style context analysis, but generic enough not to prematurely overdetermine the ontology.

They are not, by themselves, the final semantics of the trusted dependent type kernel.

10 Dependent type theory and the trusted ontology layer

The right relation between the world-model calculus and dependent type theory is subtle but promising.

10.1 What DTT can provide

A dependent type theory with contexts, substitution, Π/Σ-types, identity types, and universes provides the natural *syntax side* or internal language of the trusted ontology layer.

Categorically, this is better described via:

- categories with families (CwFs),
- locally cartesian closed categories (LCCCs),
- or related contextual/category-with-attributes structures,

not merely by saying "topos."

10.2 What DTT does not automatically provide

Dependent type theory does not by itself provide:

- an evidence algebra,
- posterior-state revision,
- provenance weighting,
- locality/gluing policies,
- or experiment-selection semantics.

So the best long-term slogan is:

> Dependent type theory can supply the internal language and context machinery of the trusted ontology/kernel layer; the world-model calculus supplies the empirical/evidential semantics layered on top of it.

10.3 The right theorem direction

The strongest future theorem pack here would show that:

1. a dependent type theory generates the relevant syntactic/contextual category,
2. a world-model hyperdoctrine interprets those contexts and substitutions,
3. substitution is respected,
4. and key type formers are interpreted by the expected semantic operations.

That would make the world-model calculus a semantic doctrine into which the trusted syntax layer is interpreted.

11 Why sheaves and subtopoi belong below the world-model layer

Locality, observers, time slices, and patchwise ontology growth are not reasons to rewrite the world-model surface. They are reasons to enrich its semantic backends.

11.1 Presheaf first, sheaf later

A presheaf backend gives local data plus restriction maps. A sheaf backend adds a gluing law: compatible local sections assemble into global sections.

This becomes useful when one has:

- observer-local semantics,
- distributed partial knowledge,
- context-sensitive concepts,
- local ontology patches that should sometimes glue.

11.2 Why crisp gluing is too strict

For AGI-style growth, exact equality on overlaps is often too strong. Local patches may only be partially compatible, evidence-weightedly compatible, or paraconsistently compatible.

So the correct long-term target is not merely ordinary sheaf gluing but:

evidence-valued or paraconsistent gluing over a locality backend.

That allows local-to-global organization without destroying useful partial syntheses.

12 Fixpoint closure and operational reasoning

The world-model calculus already has the right seed for operational reasoning: fixpoint closure.

12.1 Why fixpoints matter

Once a one-step consequence operator is available, one can define:

- immediate consequence,
- least closure,
- bounded cascades,
- eventual discovery on finite spaces,
- and later greatest/guarded variants.

This is the right semantic story for iterative reasoning, recursive closure, and stabilization.

12.2 The μ-to-ρ connection

If an operational backend such as a ρ-calculus is used for execution, the best architecture is:

- keep world-model/fixpoint semantics as the denotational spec,
- prove a μ-style query/fixpoint embedding into the operational process backend,
- show that one-step consequence and eventual stabilization correspond to operational evolution.

This is much stronger than replacing fixpoint semantics with process execution. It preserves a semantic anchor.

13 Provenance, acceptance, and trusted traces

For practical AGI use, provenance and policy-gated acceptance must become first-class.

13.1 Provenance-first semantics

A strong default story is:

- raw observations/events are wrapped with provenance,
- evidence is extracted from those assessed items,
- acceptance policies decide what enters the trusted trace,
- reasoning and treaty compliance then run over accepted traces.

This avoids the false dichotomy between "mere evidence" and "already accepted fact."

13.2 Forgetfulness should be explicit

Forgetting is not default revision. It should be a separate operation with its own laws, such as idempotence and invariance for queries outside the forgotten scope.

This is crucial for:

- privacy and redaction,
- stale-evidence expiry,
- theory contraction,
- and explicit support-scope management.

14 Experiment design and formal science

One of the most important future directions is to connect the world-model calculus to the design and evaluation of experiments.

14.1 Why this matters

A semantics of evidence is not yet a semantics of science. Science also needs:

- experiment selection,
- value of information,
- comparison of information channels,
- and policy-relative optimality.

The right long-term mathematical interfaces here likely come from categorical probability and experiment comparison (e.g. Blackwell-style orderings and Markov-category inspired transport), while the world-model calculus supplies the semantic state/evidence/query layer into which those experiment morphisms report.

15 A recommended hierarchy of world-model structures

The best long-term formulation seems to be a hierarchy, not a single monolithic class.

1. **WorldModelCore**: evidence extractor only.
2. **Generic merge layer**: possibly weak, provenance-aware merge.
3. **Overlap/correlation layer**: overlap correction or compatibility.
4. **AdditiveWorldModel**: exact additivity under independent accumulation.
5. **Forgetting layer**: explicit contraction/redaction.
6. **Locality backend**: presheaf first, sheaf/subtopos later.
7. **Closure/fixpoint layer**: consequence and stabilization.
8. **Typed semantic transport layer**: institution + hyperdoctrine.
9. **Trusted ontology/kernel bridge**: dependent type theory or other certified syntax side.

The existing additive world-model should remain the named canonical subclass, not be weakened in place.

16 A practical theorem roadmap

The following theorems would most strongly stabilize the theory.

16.1 For the additive fragment

1. **Classification theorem**: every additive generic world model on multisets is uniquely induced by its singleton observation surface.
2. **Functoriality**: additive extension is natural under additive homomorphisms of evidence carriers.
3. **No-go results**: additive plus idempotent merge plus nontriviality are incompatible; additive plus exact inverse forgetting is incompatible.
4. **Representation theorem**: additive sufficient-statistic semantics induce additive world models.

16.2 For non-additive extensions

1. **Overlap-corrected merge**: additive recovery under explicit independence.
2. **Trust action laws**: source discounts act functorially on evidence.
3. **Forgetting invariance**: queries outside forgotten scope are unaffected.
4. **Evidence-valued gluing**: local compatibility induces global assembly under policy.

16.3 For the syntax bridge

1. interpretation of substitution/reindexing,
2. interpretation of key dependent type formers,
3. reduction soundness into world-model consequence,
4. and ideally an initiality/universality theorem for the syntax-to-WM interpretation.

17 Working assumptions for future development

For future discussion and implementation work, the following assumptions should be treated as the current clean baseline.

1. The world-model calculus is primarily a semantics/evaluation interface, not a global proof system.
2. The additive fragment is mathematically distinguished and should remain explicit.
3. Strength, confidence, acceptance, and intervals are views of evidence, not the evidence itself.
4. "Actuality" should be modeled as event-occurrence evidence together with policy-relative acceptance, not as metaphysical certainty.
5. Governance reasoning benefits from event subsumption and treaty kernels, not only exact event matching.
6. Kripke and neighborhood semantics are backends or endpoints, not rivals to the core evaluator.
7. Dependent type theory is a plausible syntax/internal-language side of the trusted ontology layer, but not a replacement for evidence semantics.
8. Locality, gluing, and observer semantics belong below the world-model interface as backend refinements.
9. Fixpoint closure should remain the denotational spec for iterative reasoning even if operational backends are added.
10. Experimental design and value-of-information semantics are natural next layers for turning evidence semantics into a formal science engine.

18 Conclusion

The most robust conceptual picture is now clear.

A world-model calculus should be treated as an evidence-carrying semantic substrate with a mathematically distinguished additive fragment, a typed categorical transport layer, and multiple logical/probabilistic backends. Dependent type theory should be viewed as a candidate syntax/internal-language side of the trusted ontology layer, not as a replacement for empirical semantics. Provenance, acceptance, forgetting, locality, and experiment design should be layered on top or below the core in their proper places rather than collapsed into a single overloaded notion of "truth."

The additive fragment is special, but not absolute. It is the right semantics of independent evidence accumulation, not the unique semantics of all epistemic policy. The correct long-term architecture therefore preserves that fragment as a canonical center while generalizing around it in a disciplined hierarchy.

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